

Ravi Yadav

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[LinkedIn](#) · [GitHub](#) · [Leetcode](#) · [Portfolio](#)

SUMMARY

Computer Science & AI undergraduate with experience in machine learning, software engineering, and applied research. Second author of a physics-informed electricity demand forecasting paper presented at IEEE APPEEC 2026. Experienced in developing AI systems, fullstack products, and data-driven solutions to analyze complex problems, improve processes, and support practical decision making.

EDUCATION

Bachelor of Technology in Computer Science & Artificial Intelligence

2023 – 2027

Newton School of Technology, Rishihood University | CGPA: 7.05/10 (through Semester 6)

ACHIEVEMENTS & ACTIVITIES

- **IEEE PES Energy Shark Tank 2026 (18th APPEEC, Singapore):** FlexGrid pitch selected among the **Top 5** in the IEEE PES YP Industry Innovation Session; awarded a Certificate of Achievement for the **most outstanding innovative solution** and placed **3rd** overall.
- **1st Place, NST Startup Foundry 2026:** Won with Jarvis, an AI-powered personal intelligence system, pitched to Google Cloud and Microsoft for Startups.
- **JEE Advanced 2023:** AIR 6522 (180,000+ candidates). **JEE Main:** 97.84 percentile (1.1M+ candidates).
- **Mentorship:** Mentored **10+** students in Data Structures & Algorithms.
- **IEEE Student Member (2026):** Student Member in good standing.
- **Jaipur Under-16 Cricket Team:** Represented Jaipur as an opening batsman and part-time wicketkeeper.
- **More Than Me:** Volunteer supporting children in need.

EXPERIENCE

Software Engineer Intern : Algocept (Remote, Noida, India)

Jan 2025 – Apr 2025

- Improved a **React.js** and **Tailwind CSS** admin dashboard by resolving **20+ UI/UX and responsiveness issues**, delivering a more **consistent cross device experience**.
- **Developed** a RESTful API supporting **3 country specific configurations** using NestJS, TypeScript, and MongoDB, enabling dynamic footer content.
- **Optimized** admin dashboard loading for **500+** users by implementing pagination, **reducing** initial load time by **~50%** while limiting each request to 5 users.

RESEARCH & PUBLICATIONS

C. Murali Madhav, Ravi Yadav, K. Mehra, A. Tewary, and S. Aggarwal. “*KAT-PatchTST: Physics-Informed Forecasting with Kirchhoff Conservation*.” 18th Asia Pacific Power and Energy Engineering Conference (APPEEC), Singapore, August 2026 (IEEE; Paper ID 190).

- Accepted and presented at **IEEE APPEEC 2026**; conference proceedings publication pending.
- Stage 1 of the **Watt-IF** research project; Stages 2 and 3 under active development, targeting ICML.
- Provisional patent filed.

RESEARCH & PROJECTS

Watt-IF : Physics Informed Power Grid Forecasting ([PDF](#))

Sep 2025 – Present

- **KAT-PatchTST (Stage 1, 2nd author):** Developed a physics-informed BA-aggregate load forecasting framework combining **Channel Independent PatchTST**, **TimeXer cross attention**, **Kirchhoff conservation regularization**, and **ReLoBRaLo dynamic loss balancing**. Achieved **3.55% demand MAPE** across six BAs on the EIA930 protocol using a **168-hour context window** (30% shorter than the published 240-hour baseline) and approximately **0.6M parameters**, making the model **3–10× leaner than comparators**.
- **Stage 2 (In Development):** Extending the forecasting framework toward operational feasibility analysis over the BA interchange network to identify systemic bottlenecks and quantify node criticality from forecasted grid states.
- **Stage 3 (In Development):** Developing a conditional grid partitioning policy to minimize allocation failures under forecasted operating conditions, motivated by ORNL's reported **\$121B annual cost of major U.S. power outages in 2024**.

PrepLens : Interview Knowledge Platform ([Demo](#))

Sep 2026 - Present

- Built and deployed **PrepLens**, a full-stack interview knowledge platform (React, Node.js/Express, MongoDB) for approximately **2,000 B.Tech students** at NST, designed to help students save **10+ hours of unstructured preparation**; owned it end to end, from **database design** and REST APIs to UI, search, and a feedback-driven redesign.
- Designed a **pre-publication moderation workflow** with an admin review queue, where every submission is approved, or rejected with feedback to the author, and each decision is audited by reviewer; added company-name deduplication (alias matching with admin merge/reject) to keep search filters clean.
- Implemented **role-based access** with separate student and admin sign-in over Google OAuth, restricted to the college domain plus an admin allowlist; server-side hashed sessions and API-level anonymity protect author identity, deployed on **Vercel and Render** and backed by **117 automated tests**.

ForestLens : Tree Crown Detection & Canopy Area Estimation ([Demo](#))

Aug 2026 – Sep 2026

- Built and deployed a **geospatial ML web application** for individual **tree crown detection and canopy area estimation** from high resolution satellite imagery using **DeepForest**, **Rasterio**, and **Streamlit**.
- Diagnosed and fixed a hidden DeepForest tile scale dependency that fragmented crowns across tile sizes, improving crown geometry consistency and pipeline reproducibility.
- Investigated native-resolution detection failures through documented scale matching, reaching **71 trees/ha with 5.2m crowns** while explicitly reporting uncertainty and withholding unsupported measurements.

Optiforge : Neural Options Pricing ([GitHub](#))

Sep 2025 – Dec 2025

- Developed a hybrid options pricing system combining **LSTM based modeling** with **GARCH volatility**, benchmarked against the **Black Scholes pricing framework**.
- Built an interactive **dashboard** with heatmaps and sensitivity analysis to explore pricing behavior across model variants.

SKILLS

ML & AI: Machine Learning, Deep Learning, NLP, Computer Vision, Time Series Forecasting, Scikit-learn, TensorFlow, PyTorch

Generative AI: LLMs, RAG, LangChain, LangGraph, OpenAI API

Languages & Scientific Computing: Python, SQL, JavaScript, Java, OOP; NumPy, Pandas, Matplotlib

Databases: PostgreSQL, MySQL, MongoDB

Web, Backend & Tools: React, Next.js, Node.js, Express, REST APIs, System Design, Git/GitHub